The Twisted Transfer Operator

Where Busy Beaver meets harmonic analysis

BB(6) research program · 2026-09-08

1 The question, as a Fourier problem

A seed halts when its offset o (the counter gap) lands in the fatal set $\{1, 3, 5\}$. To detect that among the $\sim S$ possible offsets at scale S, expand the indicator in characters:

$$\mathbf{1}_{\{1,3,5\}}(o) = \tfrac{1}{M} \sum_{t} \widehat{\mathbf{1}}(t)\, \mathbf{e}\big(\tfrac{t\,o}{M}\big), \qquad \mathbf{e}(u) = e^{2\pi i u}.$$

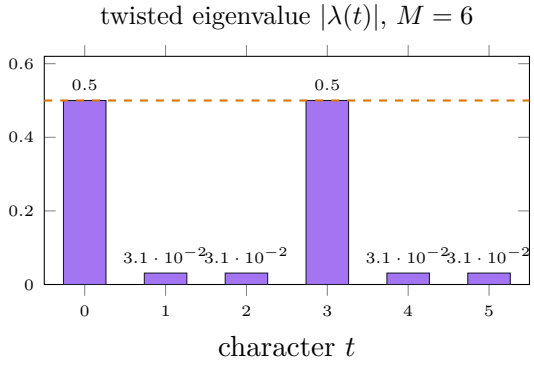
Averaging over orbits turns each term into a power of a **twisted transfer operator**

$$(\mathcal{L}_{s,t} f)(x) = \sum_{k} |g_k'(x)|^s\, \mathbf{e}\big(\tfrac{t\,o_k(x)}{M}\big)\, f(g_k x).$$

The halting hazard is controlled exactly when these operators *contract* for every non-trivial character t.

2 Aperiodicity: the characters do decay

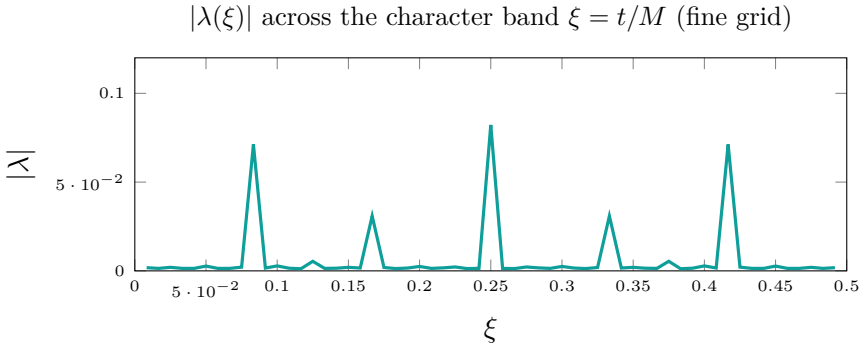
They do. Below, the modulus of the twisted leading eigenvalue $|\lambda(t)|$ for $M = 6$: the trivial character $t = 0$ sits at the top, and the *parity* character $t = 3$ (the offset is always odd — harmless) also does not decay; every **genuine** character is crushed to ≈ 0.03. This "aperiodicity" is proved two independent ways, including an exact cycle-cohomology test with explicit witnesses.



The consequence: the offset equidistributes. Mod 5, the five residues occur with frequencies $0.199, 0.200, 0.202, 0.200,$ — uniform to a fraction of a percent.

3 The uniform gap — and the open lemma

For each *fixed* M the contraction is rigorous (finite-extension spectral-gap theory over the mixing base). The subtlety is that detecting $\{1, 3, 5\}$ exactly forces the modulus M to *grow with the scale*. The plot below sweeps the character frequency $\xi = t/M$ across the band: the profile is flat and small, capped near **0.082**, with no spike toward 1 — *no resonance*.



The twisted Lasota–Yorke inequality has a variation constant that grows like $(1+|t|)$; beating that as $M \to \infty$ — turning "no resonance" into a *uniform* gap — is a **Dolgopyat oscillatory-cancellation estimate**. Our operator sits outside the settled theory on two counts: the cost is a multiplicative coordinate (not a Birkhoff sum, so Baladi–Vallée does not apply), and the branch matrices have determinant ± 3 (non-unimodular, so homogeneous-dynamics is out). This one lemma is stated precisely and *not* proved; it is the honest frontier.

What is proved: the twisted Lasota–Yorke inequality and fixed-M equidistribution. What is open: the modulus-uniform gap. The numerics establish that the gap is not blocked by a resonance — not that it is uniform.